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Identification of essential genes of *Pseudomonas aeruginosa* for its growth in airway mucus

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Pseudomonas aeruginosa **has been identified as an important causative agent of airway infection, mainly in cystic fibrosis. This disease is characterized by defective mucociliary clearance induced in part by mucus hyper-production. Mucin is a major component of airway mucus and is heavily *O*-glycosylated, with a protein backbone. Airway infection is known to be established with bacterial adhesion to mucin. However, the genes involved in mucin degradation or utilization remain elusive. In this study, we sought to provide a genetic basis of *P. aeruginosa* airway growth by identifying those genes. First, using RNASeq analyses, we compared genome-wide expression profiles of PAO1, a prototype *P. aeruginosa* laboratory strain, grown in M9-mucin (M9M) and M9-glucose (M9G) media. Additionally, a PAO1 transposon (Tn) insertion mutants library was screened for mutants defective in growth in M9M medium. One mutant with a Tn insertion in the *xcpU gene* (*PA3100*) was determined to exhibit faulty growth in M9M medium. This gene contributes to the type II secretion system, suggesting that *P. aeruginosa* uses this secretion system to produce a number of proteins to break down and assimilate the mucin molecule. Furthermore, we screened the PAO1 genome for genes with protease activity. Of 13 mutants, one with mutation in *PA3247* gene exhibited defective growth in M9M, suggesting that the *PA3247*-encoded protease plays a role in mucin utilization. Further mechanistic dissection of this particular process will reveal new drug targets, the inhibition of which could control recalcitrant *P. aeruginosa* infections.**

Keywords: *Pseudomonas aeruginosa*, airway infection, mucin, type II secretion system, protease

Introduction

Pseudomonas aeruginosa is a species of Gram-negative bacteria found in different habitats and causing mainly nosocomial infections (Gi *et al.*, 2015). It is also responsible for other kinds of infections, including urinary tract infections, burn wound infections, and infections in patients with cystic fibrosis (CF) (Porras-Gomez *et al.*, 2012). *P. aeruginosa* secretes a number of mediators during the course of infection, enabling the bacteria to evade host responses. As a consequence, significant complications emerge in the host. The virulence factors of *P. aeruginosa* include cell-associated factors like flagella (for motility), lipopolysaccharide, *lecA*, and *lecB* and extracellular factors like elastase, neuraminidase, and pyocyanin pigment (Galloway, 1991; Strateva and Mitov, 2011; van't Wout *et al.*, 2015). The bacteria form a community-like structure known as a biofilm, which is maintained by the secretion of extracellular polymeric substances that support the biofilm matrix and impart tolerance to host innate immunity mechanisms like phagocytosis. Biofilms also enable bacterial resistance to a wide range of chemicals, disinfectants, and antibiotics, enhancing survival under conditions that would otherwise hinder growth (Ma *et al.*, 2009; McConoughey *et al.*, 2014).

Airway mucus serves as a defense mechanism against invading microorganisms and toxic chemicals (Fahy and Dickey, 2010). Even though mucus secretion is essential, an imbalance in its production can lead to disease. Lung injury as a result of a deficient mucus barrier, excessive mucus secretion, or impaired clearance (as in CF) can contribute to infection, especially chronic infection with *P. aeruginosa* (Knowles and Boucher, 2002; Thornton and Sheehan, 2004). Mucus is a complex gel-like structure that is rich in host factors that provide immunity against bacteria, viruses, and fungi. The major component of mucus is a glycoprotein known as mucin (Rogan *et al.*, 2006). Mucin is either membrane-bound (like MUC 1) with a role in sensing external information and translating it into cells, or secreted, the type of which is characterized by a high molecular weight and viscosity. Mucin is a highly glycosylated molecule with a number of carbohydrates, including *N*-acetyl galactosamine, *N*-acetyl glucosamine, fucose, galactose, and sialic acid. It can also contain mannose and sulfate, and it is rich in serine and threonine residues that function as a scaffold upon which long chains of oligosaccharide are formed (Bell *et al.*, 2003; Bansil and Turner, 2006).

Because of its glycosylated structure, mucin is considered an important source of carbon, imparting bacteria with the ability to cleave glycosidic linkages, an advantage for growth (Rabiu and Gibson, 2002). Mucin degradation is accomplished by a diverse group of enzymes, including proteases, glycosidases, sialidases, and sulfatases (Hoskins and Boulding, 1981; Hoskins *et al.*, 1985; Corfield *et al.*, 1992). A number

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of bacteria that colonize the oral cavity and intestinal tract, such as Streptococcus, Helicobacter, Akkermansia, and Bacteroides, produce several mucin-degrading enzymes (Macfarlane *et al.*, 2005; Derrien *et al.*, 2014).

In this study, we investigated the capability of *P. aeruginosa* to grow using mucin as a carbon source, since it shows strong virulence and dominance over other bacterial species in a mucin-rich environment like the airway of CF patients. We used a genome-wide approach to provide a genetic basis for these virulence mechanisms. We propose new drug targets, the inhibition of which could control persistent *P. aeruginosa* infections.

Materials and Methods

Bacterial strains and growth conditions

The bacterial strain used in this study was the PAO1 strain of *P. aeruginosa* (Yoon *et al.*, 2006). Luria-Bertani medium (LB, 10 g tryptone, 5 g yeast extract, and 10 g NaCl per L) and LB agar were used to grow bacterial pre-cultures. For main bacterial cultures, M9M (M9 salt + 0.5% mucin) and M9G (M9 salt + 0.4% glucose) were used. Mucin and glucose were purchased from Sigma Corporation. Before bacterial incubation with M9 medium, LB-grown bacteria were washed using M9 salt and diluted 10,000-fold in 10 ml of M9M or M9G medium. Bacterial cells were grown for 16–20 h with shaking at 220 rpm in a 37°C incubator. Bacterial growth was assessed by measuring colony forming units (CFU) after 16 h using M9M or M9G medium or in M9 medium lacking a carbon source as a negative control.

Transposon mutagenesis

Approximately 3,000 random insertions were established in the PAO1 strain using the pBTK30 plasmid. It was first transformed into the *E. coli* SM10 λ*pir* strain (transposon vector) and then conjugated with the wild-type PAO1 strain. For counter-selection, irgasan (Sigma) and gentamycin (Sigma) were used (Filloux and Ramos, 2014). Transposon mutants were cultured in M9M and tetrazolium violet (Sigma) was added to measure bacterial growth as described previously (Sabaeifard *et al.*, 2014). Additionally, bacterial CFU were counted for confirmation.

Bacterial growth measurement by tetrazolium violet

Tetrazolium violet has been used to measure cell metabolic activity, an indication that represents bacterial growth. Several enzymes like NAD(P)H-dependent cellular oxidoreductase are capable of reducing tetrazolium violet, to the insoluble formazan with purple color, which can be detected colorimetrically (Sabaeifard *et al.*, 2014). Tetrazolium violet (0.01%) has also been used in microplates to measure bacterial proliferation in a high-throughput format (Tsukatani *et al.*, 2008). We used tetrazolium violet assay in the current study, since it was difficult to monitor bacterial growth by measuring OD_{600} values in M9M media. The presence of mucin, a high molecular weight polymer, interferes with the turbidity that proportionally increased as bacterial cell grew.

Arbitrary PCR

Genomic DNA of confirmed candidates was extracted using an extraction kit (INTRON Biotechnology). Adding arbitrary primers (Jacobs *et al.*, 2003; Liberati *et al.*, 2006) and transposon-specific primers (O'Toole and Kolter, 1998; Goodman *et al.*, 2004), a PCR reaction was performed. The PCR product was used as the template for the second PCR reaction (Filloux and Ramos, 2014). After cleaning the PCR product with a PCR clean-up kit, samples were sent for DNA sequencing (Macrogen Corporation). The sequencing data were subjected to a BLAST analysis using the NCBI website.

RNA Sequencing

A genome-wide transcriptome analysis was performed by RNASeq analysis with detailed procedures described elsewhere (Oh *et al.*, 2015). In brief, PAO1 cells pre-cultured in LB medium for 16 h were washed with M9 salts and diluted 10,000-fold, followed by subculture in 200 ml of M9M or M9G medium (as a control) overnight with 220 rpm shaking in a 37°C incubator. RNA was extracted with an RNA extraction kit (Qiagen Corporation) and sent for sequencing at Chun Lab, Inc. RNA sequence data were analyzed using software from Chun Lab, Inc.

Quantitative real-time PCR

Target gene transcript levels were quantified using qRT-PCR to verify RNA sequencing results, as described previously (Yoon *et al.*, 2011). The bacterial cells were cultured in LB medium for 6 h and sub-cultured in M9M or M9G medium overnight, followed by RNA extraction. 16s ribosomal RNA gene transcript levels were measured and used to normalize the target genes.

Elastase assay

Elastase activity was measured from PAO1 and mutant cells using elastin Congo red (Sigma) after growth overnight at 37°C, as described previously (Senturk *et al.*, 2012).

Statistical analysis

The experiments were repeated for reproducibility and the mean ± standard deviation (SD) were calculated. For data analysis, Student's unpaired *t*-test and ANOVA were performed. *P*-values were considered statistically significant if they were < 0.05.

Results

P. aeruginosa can use mucin as a carbon source to support its growth

The PAO1 genome spans approximately 6.3 Mbp and includes a number of genes and operons that are involved in substrate metabolism and breakdown, imparting the bacteria with the ability to survive in different environments (Frimmersdorf *et al.*, 2010). We sought to determine whether *P. aeruginosa* can utilize mucin to support its growth since this bacterial species has the ability to cause airway infections,

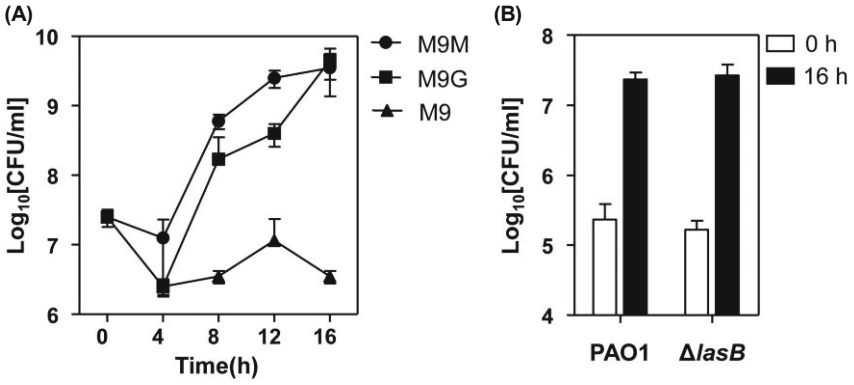


Fig. 1. Growth of PAO1 in M9 minimal medium supplemented with mucin. (A) PAO1 growth curve in M9 medium with mucin as a sole carbon source (M9M), M9 with glucose (M9G), and M9 medium without a carbon source (M9) for 16 h in a shaking incubator at 37°C. The viable count assay was used in order to determine the colony forming units CFU/ml. (B) Growth of PAO1 and its isogenic *lasB* mutant in M9M medium for 16 h was performed to test the ability of an elastase mutant to grow in M9M.

particularly in CF patients in which mucus over-secretion is reported (Davies, 2002). We used M9 minimal medium supplemented with mucin as a sole carbon source. We grew bacteria overnight at 37°C. By measuring bacterial CFU, we found that there was an approximate >100-fold increase in bacterial number, indicating that PAO1 can degrade mucin and use it for growth (Fig. 1A).

P. aeruginosa produces a number of virulence factors to evade the host response (van't Wout *et al.*, 2015). Elastase, a major zinc metalloproteinase encoded by the *lasB* gene, has been shown to degrade both innate and adaptive immune components (Kuang *et al.*, 2011; Gi *et al.*, 2014). Correspondingly, we tested the role of elastase in mucin degradation using a *lasB* mutant, monitoring its growth in M9M medium. We found similar growth between the *lasB* mutant and PAO1 after incubation in M9M medium, indicating that PAO1 uses other gene products to cleave the mucin molecule (Fig. 1B).

Genome-wide expression analysis to identify PAO1 genes essential for mucin utilization

To understand the molecular basis of this process, we conducted RNASeq analysis. Table 1 shows a list of genes that were highly expressed after incubation in M9M vs. M9G medium (each was at least 100-fold increased), indicating that growth in M9M changed the bacterial expression profile. Among those genes were two (PA0122 and PA2939) that encode a probable haemolysin and a probable amino peptidase, respectively. We then assessed mutants with defects in those two genes, comparing them to PAO1 for growth in M9M medium. We used transposon insertion mutants obtained from a *P. aeruginosa* mutant library (University of Washington). All of the tested mutants showed similar growth to PAO1, suggesting that the bacteria use other gene products to break and consume the mucin molecule. As a result, we searched the PAO1 genome database for genes with glycosi-

Table 1. List of genes with more than a 100-fold increase in expression after bacterial incubation in M9M medium, using RNASeq analysis
Expression of each gene was assessed by measuring RPKM (Reads per kilobase million).

Gene name	Locus No.	RPKM in M9G	RPKM in M9M	Function
rahU	PA0122	86.09	9040.4404	Hypothetical protein (hemolysin)
-	PA2939	17.19	2732.25	Probable aminopeptidase
lecB	PA3361	8.61	976.51	Fucose-binding lectin
phzG1	PA4216	0.2	28.65	Probable pyridoxamine 5'-phosphate oxidase
flp	PA4306	25.25	12372.410	Type IVb pilin, Flp
pra	PA4590	44.46	8372.3496	Protein activator

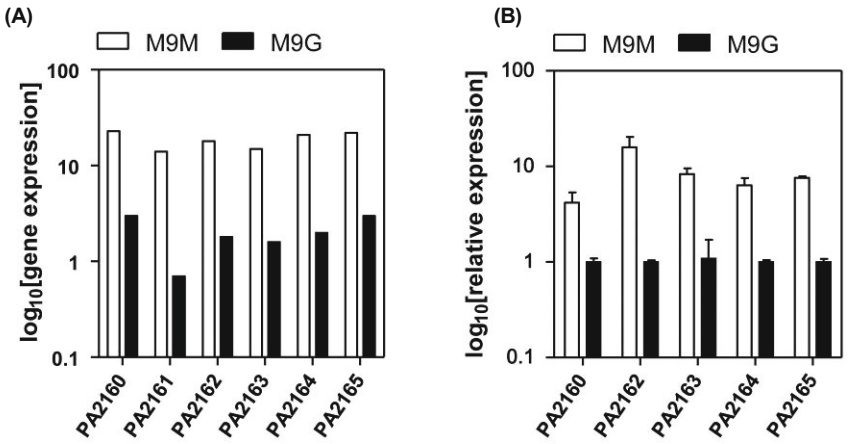


Fig. 2. PAO1 glycoside hydrolase gene expression profile after incubation in M9M. (A) Glycoside hydrolase operon gene expression in M9M medium using RNASeq. PAO1 was grown in M9M (white) and M9G (black) and bacterial pellets were processed for RNASeq analysis. (B) Quantitative RT-PCR analysis of the glycoside hydrolase operon. qRT-PCR was conducted on cDNA synthesized from 2 μg of total RNA extracted from PAO1 grown in M9M (white) or in M9G (black). Bacterial cells were harvested after 8 h of growth and subjected to RNA isolation. Transcript levels of the tested genes were normalized with 16S rRNA gene transcripts.

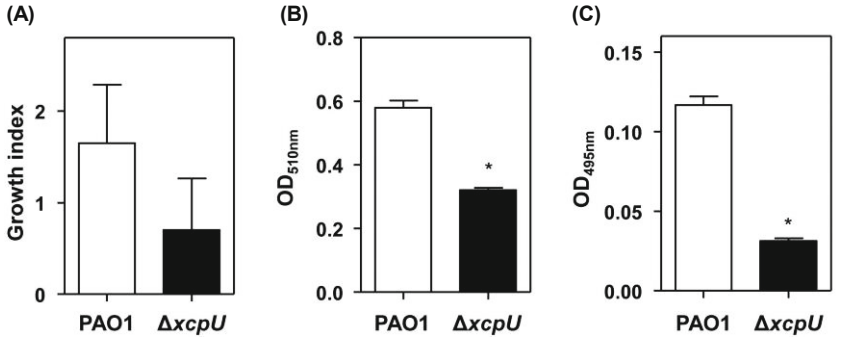


Fig. 3. Transposon mutagenesis screening. (A) A PAO1 mutant with a defect in growth in M9M medium was tested using the following growth index. Growth index = $\log_{10}$ [CFU $_{\text{after 16 h}}$ /CFU $_{\text{at initial inoculum}}$]. (B) Bacterial growth was examined using tetrazolium violet (TV) dye. (C) Detection of type II secretion system efficiency by measuring elastase secretion in the *xcpU* mutant. **P* < 0.05 vs. values of PAO1 as determined by Student's *t*-test.

dase activity. Using bioinformatics tools like BLAST analysis, we compiled a list of genes in one operon (from PA2160 to PA2165) with probable glycoside hydrolase function. We examined the expression levels of those genes using our RNA-Seq data, which illustrated high expression levels in M9M medium (an approximate 10-fold increase) (Fig. 2A). We confirmed these data using qPCR analysis (Fig. 2B). Although the genes with a probable glycoside hydrolase function showed elevated expression, their mutants showed no defect in growth compared to wild type PAO1 (data not shown).

The *PA3100* gene is essential to support bacterial growth in M9M

To identify how the PAO1 strain breaks down and assimilates mucin, we carried out a transposon mutagenesis experiment. Approximately 3,000 transposon insertion mutants were generated. A number of auxotrophic mutants were identified to show growth defects in M9M medium (namely mutants deficient in tryptophan- or leucine-biosynthesis pathway). These mutants, however, were also defective in M9G media (data not shown). Therefore, we didn't pursue any further investigation on this particular group of mutants. Of importance, one mutant exhibited a significant growth defect only when grown in M9M media. The transposon insertion was determined to occur in the gene *PA3100* (*xcpU*) using arbitrary PCR followed by sequencing (Macrogen Corporation). This gene functions in the type II secretion system, and it is annotated as being part of the *PA3095-PA3103* operon (Bleves *et al.*, 2010). Subsequent experiments showed that the *PA3100* mutant (Δ*xcpU*) grows well in LB and M9G media. However, it exhibited a decrease in growth in M9M medium, with bacterial survival measured using growth index (Fig. 3A) and tetrazolium dye (Fig. 3B).

To demonstrate that the Δ*xcpU* mutant displays abnormal protein secretion, we measured elastase quantity using the Congo red method (Senturk *et al.*, 2012). This protein is known to be secreted by the type II secretion system (Braun *et al.*, 1998). Consequently, this mutant showed a significant defect in elastase production compared to PAO1, implying atypical protein secretion (Fig. 3C). Our results indicate that PAO1 releases other proteins rather than elastase by type II secretion to break down mucin.

To confirm that a defect in the type II secretion system had an impact on bacterial growth in M9M medium, we investigated bacterial expression profiles for some of the genes involved in type II secretion, such as *xcpQ* and *xcpZ* (PA3105 and PA3095, respectively). PAO1 mutants defective in each of these two genes also exhibited compromised growth in M9M (Fig. 4A). The degree of growth defects was similar to that observed in Δ*xcpU* mutant. Again, these two mutants

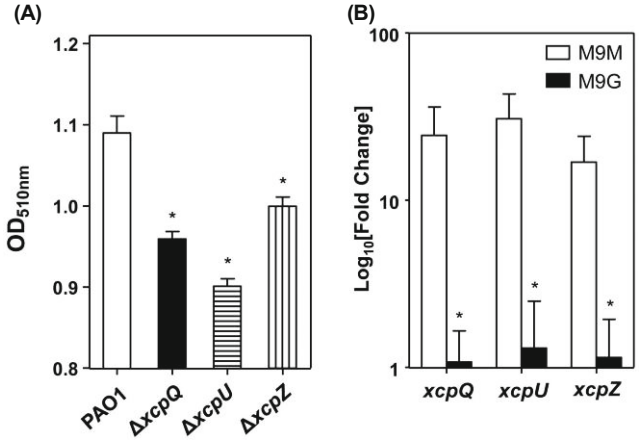


Fig. 4. Growth of type II secretion system mutants in M9M medium. (A) Growth of PAO1 and its isogenic Δ*xcpU*, Δ*xcpQ*, and Δ*xcpZ* mutants in M9M medium, measured using tetrazolium dye. (B) Expression profiles of type II secretion system genes using qRT-PCR. Experimental conditions were identical to those described in Fig. 2B.

Table 2. List of tested mutants (Tn-inserted mutants) with defects in protease activity

All of these mutants were scrutinized for proliferation in M9M medium for 16 h. A mutant of *PA3247* gene was only found to be defective in growth in M9M media. All other mutants grew normally in M9M.

Locus No.	Gene	Function
PA0766	*mucD*	serine protease MucD precursor
PA3535	-	probable serine protease
PA3257	*algO*	serine-type peptidase activity
PA5134	*ctpA*	carboxyl-terminal processing protease
PA1832	-	probable protease
PA0779	*asrA*	ATP-dependent peptidase activity
PA1801	*clpP*	ATP-dependent Clp protease
PA4171	-	probable protease
PA0355	*pfpI*	protease PfpI
PA0372	-	probable zinc protease
PA1327	-	probable protease
PA5474	-	probable metalloprotease
PA3247	-	putative aminopeptidase

grew normally in M9G (data not shown). Quantitative RT-PCR analysis showed that expression of these genes was increased >10-fold when grown in M9M vs. M9G (Fig. 4B). This led us to hypothesize that the type II secretion system is essential for bacterial growth in medium supplemented with mucin.

Screening of the *P. aeruginosa* genome for genes with potential protease activity

Mucin is a large glycoprotein with a protein core that constitutes 20% of the total polymer. The amino acids either exist in a tandem repeat (serine, threonine, and proline) or help in dimerization via disulfide bonds, as in cysteine-rich areas (Turner *et al.*, 1999; Bell *et al.*, 2001, 2003). We next hypothesized that *P. aeruginosa* produces proteases that have direct action in degrading mucin. To test our hypothesis, we searched the PAO1 genome database for genes with potential protease activity. We screened a number of Tn-inserted mutants for growth in M9M medium (Table 2). Of 13 mutants, we observed one with minimal growth in mucin-containing media using a phenotype assay and growth index values (Fig. 5). This mutant harbors a mutation in *PA3247* gene encoding a protein with probable amino peptidase activity. We concluded from these data that PAO1 may use the *PA3247* gene product to disrupt the amino acids within the mucin molecule in order to free polymer oligosaccharides and use them to support its growth.

Discussion

P. aeruginosa can remodel its metabolic processes according to the surrounding environment. This metabolic versatility greatly contributes to bacterial pathogenicity, especially in respiratory infections like those in CF patients (Son *et al.*, 2007; Hoboth *et al.*, 2009). Unlike other bacteria, *P. aeruginosa* shows a significant capability to cause chronic infection, even though the lung excretes a large quantity of mucus. This secreted material acts as a defense mechanism through which the host can dispose of the invading organisms (Girod *et al.*, 1992; Yoon and Hassett, 2004; Fahy and Dickey, 2010). Here, we tried to determine how *P. aeruginosa* causes infection in its host airway. Previous studies have emphasized that *P. aeruginosa* can grow in complex medium containing mucus as one of the components (Palmer *et al.*, 2005; Son *et al.*, 2007; Fung *et al.*, 2010). In our study, we used mucin, a large complex that gives mucus its elasticity. Because it is composed of large complexed oligosaccharides, we tested bacterial growth using M9 minimal medium supplemented with mucin as a sole source of carbon. Our data showed that PAO1 (a virulent type) can utilize mucin as an energy source.

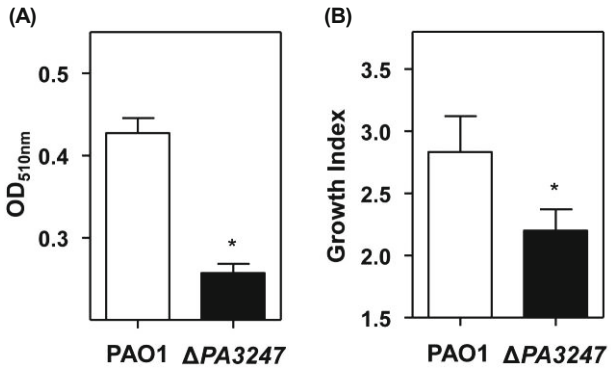


Fig. 5. Defective growth of a *PA3247* mutant in M9M medium. (A) The bacterial growth assay using tetrazolium dye. PAO1 (white) and Δ*PA3247* mutant (black) were grown in M9M and their growth was assessed by measuring $OD_{510\ nm}$. (B) The viable bacterial assay measuring CFU after bacterial incubation in M9M medium for 16 h in a shaking incubator at 37°C. **P* < 0.05 *vs.* growth of PAO1 in M9M medium, as determined by Student's *t*-test.

To explore the mechanism through which mucin degradation takes place, we focused on PAO1 up-regulated genes after bacterial incubation in M9M medium, using RNASeq data. The expression of *PA0122* and *PA2939* was highly induced (an approximate 100-fold increase), with probable haemolysin and amino peptidase functions, respectively. Those two gene mutants would have been further screened had they displayed lower growth rates in M9M medium, but our data indicated no defects in their growth. Additionally, the gut micro biome has been reported to produce a number of glycosidase enzymes that allow bacteria to digest the mucin polymer (Pettipher and Latham, 1979; Berg, 1981; Macfarlane *et al.*, 1998). We scanned the *P. aeruginosa* genome for similar genes using bioinformatics tools like BLAST analysis. One operon (*PA2160-PA2165*) with probable glycoside hydrolase function was found and was highly up-regulated after bacterial culture in M9M medium (an approximate 10-fold increase). Similarly, these gene mutants showed no impairment in growth. Based on this finding, we supposed that *P. aeruginosa* might use other genes to disrupt and transport the mucin oligosaccharide inside the cell.

We screened the *P. aeruginosa* genome for more genes involved in bacterial survival in M9M medium, using transposon mutagenesis. *PA3100*, a type II secretion system gene, was indicated as essential for bacterial proliferation in M9M. Accordingly, we tested our mutant for protein secretion using an elastase assay, since large numbers of *P. aeruginosa* proteins, including the elastase protein, are secreted through the type II secretion system (Braun *et al.*, 1998; Martínez *et al.*, 1999; Folders *et al.*, 2000; Voulhoux *et al.*, 2001; Fox *et al.*, 2008). Our examination of this mutant demonstrated abnormal protein secretion. Consequently, we scrutinized certain other mutants with defects in type II secretion system genes for growth in M9M medium; consistent with our earlier findings, we observed defects in their survival. We concluded that this secretion system is important for *P. aeruginosa* mucin utilization from the surrounding environment. Whether other secretion systems also contribute to mucin degradation remains to be determined.

Because the mucin structure contains a protein backbone, we next examined whether *P. aeruginosa* proteases had any direct effect on mucin degradation. We screened the *P. aeruginosa* genome database for genes with protease activity. Following that, we tested 13 Tn insertion mutants. We observed one mutant with an obvious defect in growth in M9M medium. It was a hypothetical protein (PA3247) with probable amino peptidase function. A previous study has illus-

trated that mucin molecules can be degraded by synthetic human neutrophil elastase (HNE) and by *P. aeruginosa* elastase B (pseudolysin) (Henke *et al.*, 2011). Even though our previous data showed that PAO1 elastase has no effect on mucin breakdown, it is possible that *P. aeruginosa* proteases like PA3247 contribute to mucin degradation and subsequently be important for bacterial proliferation inside the host. For these reasons, protease inhibitors would be useful to suppress bacterial growth.

In conclusion, our data showed that *P. aeruginosa* can proliferate in minimal medium supplemented with mucin as a carbon source. Our results provide insight into the molecular mechanisms of mucin breakdown. This study suggests that the type II secretion system is crucial for bacterial growth in M9M medium. Whether the *PA3247* gene is secreted by the type II secretion system remains to be determined in a future investigation. Our findings also suggest that inhibition of the *xcpU* gene function (and other genes from the type II secretion system) by blocking its biological activity could be a promising strategy to treat *P. aeruginosa* infections.

Acknowledgements

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